Harshal Parekh

+91 8238482880 | harshalparekh17@gmail.com | linkedin.com/in/harshal-p-parekh | github.com/harshal4172005 | Portfolio
Surat, India | Open to relocation | Available to join immediately

Summary

AI Engineer and 2026 Computer Engineering graduate. Built diffusion-based image and video generation pipelines for 5+ garment and jewellery brands at NexAI Labs, deployed as FastAPI and Docker services. Also shipped a full-stack multi-branch ERP for gaming cafés using AI coding agents. Seeking AI/ML Engineer roles.

Experience

Junior AI Engineer Nov. 2025 – Apr. 2026
NexAI Labs *Surat, India*

- Built diffusion-model image generation pipelines for 5+ garment and jewellery brands, cutting manual design effort by 60%.
- Automated image generation and content creation end-to-end with a multi-agent workflow in n8n, integrating model APIs into a single pipeline.
- Extended the pipeline to AI video generation, speeding up client content delivery by 40%.
- Packaged and deployed the AI services as REST APIs with FastAPI and Docker, serving internal teams and client-facing applications.

AI Engineer Intern Aug. 2025 – Oct. 2025
NexAI Labs *Surat, India*

- Developed prompt workflows and API integrations for the team's AI automation pipelines.
- Deployed and tested FastAPI-based ML applications on Google Cloud.

Projects

Multi-Branch Gaming Café ERP | *.NET 8, React, PostgreSQL, SignalR, Redis, Docker, Oracle Cloud* GitHub

- Built and deployed a full-stack ERP for gaming cafés, designed to manage multiple branches and 100+ gaming PCs.
- Built an ASP.NET Core backend with Clean Architecture (30+ controllers, EF Core, PostgreSQL) and a 16-module React frontend.
- Synced sessions, billing and orders in real time via 10 SignalR hubs, plus a Windows agent that locks and unlocks PCs remotely.
- Automated usage-based billing with split payments (cash, UPI, wallet), member wallets and tamper-proof daily reports.
- Added JWT auth with role-based access; deployed 7 Docker services on Oracle Cloud with Nginx, SSL and daily backups.

Multimodal RAG Research Assistant | *Python, FastAPI, Gemini, Qdrant, PyMuPDF, Groq, Docker* GitHub

- Built a multimodal RAG assistant that answers questions on 7 landmark AI papers (Transformer, BERT, GPT-3, LoRA) with page citations.
- Captioned every figure with Gemini Vision at ingestion, including vector-drawn diagrams most PDF tools miss, so figures are searchable in Qdrant.
- Added a model fallback chain (Gemini, Groq, NVIDIA NIM) and exponential-backoff retries so rate limits never block an answer.

AI CCTV Safety Surveillance | *Python, YOLOv8, PyTorch, OpenCV, Streamlit, SQLite* GitHub

- Fine-tuned YOLOv8 on a 10-class PPE dataset to detect missing hard hats, masks and vests in images and live video.
- Logged every violation with a timestamp and cropped snapshot, with a per-class cooldown to stop duplicate alerts.
- Built a Streamlit app with live WebRTC streaming, Plotly analytics, SQLite user login and PDF/CSV violation reports.

Technical Skills

Languages: Python, SQL, Java (basic)
AI / ML: Diffusion Models, YOLOv8, OpenCV, PyTorch, Scikit-learn, Pandas, NumPy, RAG, Qdrant, Gemini API
Automation: n8n, Multi-Agent Workflows, API Integrations
Backend & APIs: FastAPI, Flask, REST APIs, JSON, Pydantic, JWT, Swagger
DevOps & Cloud: Docker, CI/CD, Nginx, Linux, Git, GitHub, AWS (EC2, S3), Google Cloud, Oracle Cloud, Cloudflare R2
Databases: PostgreSQL, SQLite
Frontend & Tools: React, Tailwind CSS, HTML, CSS, Streamlit, Jupyter Notebook

Education

C.K. Pithawala College of Engineering and Technology Surat, India
B.Tech in Computer Engineering – CGPA: 8.41 / 10 *Aug. 2022 – Jun. 2026*

Certifications

Ethical Hacking & Penetration Testing – C-DAC Noida (training + virtual internship)
AWS Cloud Practitioner (CLF-C02) Exam Prep – Udemy
Python, Data Analysis & AI Foundation – SAP Code Unnati (Edunet Foundation)